



Figure 4: Cricket image

```
DISCOVERY_URL = ('https://[api].googleapis.com/'
                 '$discovery/rest?version=[apiVersion]')

def main(photo_file):
    credentials = GoogleCredentials.get_application_default()
    service = discovery.build('vision',
                              'v1', credentials=credentials)
    with open(photo_file, 'rb') as image:
        image_content = base64.b64encode(image.read())
        service_request = service.images().annotate(body={
            'requests': [{
                'image': {
                    'content': image_content.decode('UTF-8')
                },
                'features': [{
                    'type': 'LABEL_DETECTION',
                    'maxResults': 5
                }]
            }]
        })
    response = service_request.execute()
    print(json.dumps(response['responses'], indent=4))

if __name__ == '__main__':
    parser = argparse.ArgumentParser()
    parser.add_argument('image_file', help='The image you\'d
    like to label.')
    args = parser.parse_args()
    main(args.image_file)
```

Let us understand the code first. In the main block, we read the file name that we provided. The first step in the main function is to establish the credentials; we use the standard application credentials, which were set via the environment variable. We then read the file bytes and invoke the REST API endpoint for the cloud vision API. Notice that in the REST API call, we provide not just the bytes for the image but also the feature that we are interested in, i.e., `LABEL_DETECTION`, and command it to return us a maximum of five such label results. We finally invoke the API and print out the results.

The output can be downloaded from http://opensourceforu.com/article_source_code/dec16/gml.zip. Notice that each of the labels that it detected has a confidence score out of 10. It is generally considered a positive response if it is about 70-80 per cent, but it is entirely up to your application to accept these results. Note that it has also

detected erroneously that there is a baseball player—it is important to keep these points in mind.

Natural Language API

The Natural Language API helps us to identify parts of speech in a text, entities (persons, monuments, organisations, etc) and also do sentiment analysis. In the specific example that we will execute here, we are going to test how the API extracts entities. This will help us to identify the different entities that were present in the piece of text that we provided to it.

The text that we will provide to the REST API is as follows:

The search for a new chairman of Tata Sons following the ouster of Cyrus Mistry has thrown up quite a few names. The high-profile names include Indra Nooyi, the head of PepsiCo Inc, Arun Sarin, the former head of Vodafone Group, and Noel Tata, chairman of the Tata retail unit, Trent.

The code is shown below and it follows a pattern similar to the vision API that we saw earlier.

```
from googleapiclient import discovery
import httplib2
import json
from oauth2client.client import GoogleCredentials

DISCOVERY_URL = ('https://[api].googleapis.com/'
                 '$discovery/rest?version=[apiVersion]')

def main():
    http = httplib2.Http()

    credentials = GoogleCredentials.get_application_default().
    create_scoped(
        ['https://www.googleapis.com/auth/cloud-platform'])

    http=httplib2.Http()
    credentials.authorize(http)

    service = discovery.build('language', 'v1beta1',
                              http=http,
                              discoveryServiceUrl=DISCOVERY_URL)

    service_request = service.documents().analyzeEntities(
        body={
            'document': {
                'type': 'PLAIN_TEXT',
                'content': "The search for a new chairman of Tata
                Sons following the ouster of Cyrus Mistry has thrown up quite
                a few names. The high-profile names include Indra Nooyi, the
                head of PepsiCo Inc, Arun Sarin, the former head of Vodafone
                Group, and Noel Tata, chairman of the Tata retail unit
                Trent."
            }
        })
```